# Mohammad Yak Safu

mohammadyaksafu@gmail.com • +880 1642-468704 • Gazipur, Dhaka, Bangladesh
LinkedIn • GitHub • Kaggle • LeetCode • Codeforces • Portfolio

## Education

**Shahjalal University of Science and Technology (SUST)** *2023 – Present*
B.Sc. in Software Engineering Sylhet, Bangladesh

## Research & Publications

- **COMPAS 2026 (IEEE) – Accepted:** "An Empirical Study of Detecting AI-Generated Text" – benchmarked transformer and gradient-boosting classifiers for machine-generated text detection.
- **ICCIT 2025:** "EquiTea: An IoT–ML–Blockchain Integrated System for Transparent Wage Payments" [Paper]

## Competitions

- **AI Dev Fest 2026 (DIU) – AI Hackathon: Champion (1st Place)** [Results]
- **AI Dev Fest 2026 (DIU) – Datathon: 2nd Runner-up (3rd Place)** [Results]
- **DUET AI Datathon (Kaggle: Road Vision) – 9th Place:** computer-vision pipeline for road-scene recognition. [Leaderboard]
- **Bengali LLM Hallucination Detection (Kaggle, invitation-only):** 84th of 222 teams [Competition]

## Technical Skills

**Languages:** Python, C++, Java, C, SQL, JavaScript/TypeScript
**ML & Deep Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost, CatBoost, Transformers, Sentence Transformers, LoRA/QLoRA fine-tuning
**Computer Vision & NLP:** YOLO, VLMs, RAG pipelines, LLM application development
**Data & MLOps:** NumPy, Pandas, FastAPI, ChromaDB, Docker, Git/GitHub, Linux

## Projects

**Shurokkha – Real-Time Fraud & Scam Detection Copilot** *LightGBM, SHAP, FastAPI, Gemini, Docker*
*Champion, AI Dev Fest 2026 (DIU CPC × upay AI Hackathon)* **[Link]**

- Built a calibrated LightGBM risk scorer over 46 point-in-time behavioural, device and SIM-swap features: **PR-AUC 0.992** and **99.1% recall at 1% FPR**, served in ∼5 ms per transfer.
- Added verified TreeSHAP reason codes in Bangla/English, Louvain + PageRank graph detection of money-mule rings, and an evidence-grounded RAG case-summary copilot for analysts.

**DRISHTI AI – Crowdsourced Road-Hazard Intelligence** *Ultralytics YOLO, FastAPI, OpenCV, MongoDB, React*
*Geospatial road-condition platform for Bangladesh* **[Link]**

- Trained and deployed a six-class hazard detector (pothole, crack, open manhole, waterlogging, debris, broken road) with an asynchronous inference worker and bounding-box annotation pipeline.
- Designed a severity engine that scores real danger (0–100) from hazard area, hazard class and detector confidence, driving a worst-first repair priority queue for authorities.

**ThreadLight – Compliance Evidence Platform** *LangGraph, Gemini, Llama 3.2, pgvector, Presidio, FastAPI*
*Document AI pipeline for garment-export audit readiness* **[Link]**

- Built a LangGraph pipeline with human-in-the-loop interrupts: Docling/Tesseract OCR (EN+BN), Presidio PII redaction, local Llama 3.2 classification with Gemini fallback, and schema-constrained field extraction validated against the source quote.
- Implemented hybrid retrieval (pgvector + full text, HyDE rewriting, reranking) for clause mapping, plus an MCP server and automated evaluation suites for extraction and retrieval quality.

**Also:** *PDF RAG Application* – FastAPI + ChromaDB retrieval service with semantic search and grounded LLM answers **[Link]**

## Teaching & Leadership

**Director of Robotics – RoboSUST** (R&D Secretary, 2025–2026) *2026 – Present*
**Instructor, 8-Day Machine Learning Workshop – RoboSUST** *2026*
**Organizer & Instructor, Free Python & ML Bootcamp (400+ students)** *2025*

## Problem Solving

- **Codeforces** (Kn0wthyself): Pupil, rating 1263 (max. 1282) – 178 problems solved
- **LeetCode** (mohammadyaksafu): 184 problems solved (C++, Java)